It is probably concerned in optic reflexes **MidBrain Activation** . The metencephalon and myelencephalon together compose the rhombencephalon which divides into pons and medulla oblongata . These terms relate to the front-to-back arrangement of structures in the developing embryo although during the course of development the positions change so that the forebrain is essentially sitting on top of the **Midbrain** and hindbrain. The substantia nigra is a group of dark-colored dopaminergic cells. The corpus callosum plays an important role in transferring information from one hemisphere to the other **MidBrain Activation** . It is involved in the coordination of motor movements as well as basic facets of memory and learning. The principal gray masses of the tegmentum are the red nucleus and the interpeduncular ganglion; of its fibers the chief longitudinal tracts are the superior peduncle the medial longitudinal fasciculus and the lemniscus. For descriptive purposes each cerebral hemisphere can be divided into six lobes. The substantia nigra plays an important role in movement learning and addiction. Located above the brainstem the thalamus processes and transmits movement and sensory information.

Like the spinal cord the brain is made of mainly gray matter and white matter arranged in distinct layers **MidBrain Activation** . The sagittal section is also parallel to the vertical plane but a midsagittal section would divide the head into right and left halves. The tegmentum sits beneath the tectum. The circle of Willis ensures a continuous blood supply to the brain. There are subcortical areas in the telencephalon like the basal ganglia and the amygdaloid nucleus complex **MidBrain Activation For Adults** . The efferent projections include cortico bulbar cortex to brain stem and cortico spinal cortex to spinal cord axons. 7B is a section taken at the level of the upper medulla. The right brain controls the left side of the body. Bechterew is of the opinion that the fibers from the motor area of the cerebral cortex form synapses with cells whose axons pass to the motor nucleus of the trigeminal nerve and serve for the coordination of the muscles of mastication. The pons connects the medulla to the cerebellum and helps coordinate movement on each side of the body.

It is essentially a relay station taking in sensory information and then passing it on to the cerebral cortex. During development the **Midbrain** forms from the middle of three vesicles that arise from the neural tube. According to Edinger it extends to the so-called nucleus of the posterior longitudinal bundle in the hypothalamic region but this is uncertain and the fibers above the nucleus of the oculomotor are smaller in diameter than the rest of the bundle. Below the telencephalon afferent nerve fibers from each eye decussate in the optic chiasm and join uncrossed fibers to form the optic tract. Landmark structures include the fourth ventricle the pons tegmentum which includes the abducens nuclei; the pons base which includes the corticofugal fibers and pontine nuclei; and the middle cerebellar peduncles **MidBrain Activation For Adults** . Instead the goal of this brain tour is to familiarize you with major brain structures and their functions. VII and VIII also arise from the medulla and V has branches in the **Midbrain** and medulla. The frontal lobe is the portion which is rostral to the central sulcus and above the lateral fissure and it occupies the anterior one third of the hemispheres . There are 86 main nerves branching off from the CNS 12 pairs of cranial nerves that branch off from the brain and 31 pairs of spinal nerves that branch off from the spinal cord. These three vesicles further differentiate into five subdivisions: telencephalon diencephalon mesencephalon metencephalon and the myelencephalon .

The basal ganglia works along with the motor areas of the cortex and cerebellum for planning and coordinating certain voluntary movements. It is also the origin for 10 of the 12 cranial nerves. At this level a landmark structure of the diencephalon is the thalamus which surrounds the third ventricle. Transverse Section through the Spinal Cord. The medial lemniscus may be considered as the upward continuation of the posterior funiculus of the spinal cord and to convey conscious impulses of muscle sense and tactile discrimination. They arise from the cerebral hemispheres I and II from the **Midbrain** III and IV from the pons V VI VII and VIII and from the medulla IX X XI and XII. Interneurons of the reticular formation receive some of the cortico-bulbar fibers from the motor cortex. This is because we move our hands and lips all the time and both are very sensitive. The two vertebral arteries supply blood to the back of the brain **MidBrain Activation For Adults** . The basal ganglia is found in the white matter.

The corpus callosum is a collection of nerve fibers that connect the two hemispheres **MidBrain Activation** . White nerve fibers underneath carry signals between the nerve cells and other parts of the brain and body. The non-dominant parietal lobe processes visual-spatial information and controls spatial orientation **MidBrain Activation For Adults** . On its medial surface there is a prominent fissure the calcarine fissure and parieto-occipital sulcus. The cerebrum is divided in to left and right hemispheres. We will discuss the relevance of the degree of cortical folding or gyrencephalization later. They are usually the result of a cardioembolism a blood vessel obstruction within or around the heart muscle. These nerve cells are important in regulating motor function and emotion and cell death leads to symptoms such as tremors physical instability and emotional changes. The nuclei of the 3rd the 4th and part of the 5th are located in the **Midbrain** tegmentum. Blood vessels located on the surface of the brain and the spinal cord are found on top of the pia matter.

The third ventricle is beneath the corpus callosum and surrounded by the thalamus. Scattered throughout the central gray stratum are numerous nerve cells of various sizes interlaced by a net-work of fine fibers. The **Midbrain** controls many important functions such as the visual and auditory systems as well as eye movement. It consists of the amygdala the septum and the hippocampus. The horizontal section is parallel to the horizontal plane and a mid-horizontal section would divide the head into superior and inferior halves. The tegmentum is continuous below with the reticular formation of the pons and like it consists of longitudinal and transverse fibers together with a considerable amount of gray substance. The **Midbrain** is the smallest region of the brain that acts as a sort of relay station for auditory and visual information. The secretion of melatonin increases in the dark and decreases when it is bright. Finally the forebrain is the most recent evolutionary addition and is the last to develop prenatally. The two last-named strata are sometimes termed the gray-white layers from the fact that they consist of both gray and white substance.

It continues from the cerebrum above and connects to the spinal cord below **MidBrain Activation For Adults** . The surface of the cortex consists of unmyelinated gray neurons which is why this part of the brain is sometimes referred to as gray matter. This system is an interconnected series of spaces within the brain which contains the CSF . Lateral view of human embryo at the beginning of the 3rd and 5th week of gestation. Having trouble remembering which bits of the brain do what? The following two videos show how you can use visual strategies to link structures and functions1. The tectum is the dorsal or roof part of the **Midbrain** and controls visual and auditory reflexes. The occipital lobe is the most caudal part of the brain lies on the tentorium cerebelli and is comprised of several irregular lateral gyri. Most of these fibers are crossed but some are uncrossed. The caudal part of the superior temporal gyrus which extends up to the parietal cortex forms part of Wernickes area **MidBrain Activation For Adults** . The red nucleus is situated in the anterior part of the tegmentum and is continued upward into the posterior part of the subthalamic region.